

A CFD-based surrogate model for predicting flow parameters in a ventilated room using sensor readings

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ABSTRACT

In this work, we develop a computational fluid dynamics (CFD)-based surrogate model, which predicts flow parameters under different geometrical configurations and boundary conditions in a benchmark case of a mechanically ventilated room with mixed convection. The model inputs are the temperature and velocity values in different locations, which act as a surrogate of the sensor readings. The model's output is a set of comfort-related flow parameters, such as the average Nusselt number on the hot wall, jet separation point, average kinetic energy, average enstrophy, and average temperature. We tested four different machine learning methods, among which we chose the gradient boosting regression due to its accurate performance. We also adapted the developed model for indoor environment control applications by determining the optimal combinations of sensor positions which minimize the prediction error. This model does not require the repetition of CFD simulations in order to be applied since the structure of the input data imitates sensor readings. Furthermore, the low computational cost of the model execution and good accuracy makes it an effective alternative to CFD for applications where rapid predictions of complex flow configurations are required, such as model predictive control.

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1. Introduction

The quality of the indoor environment is an important issue in our daily life, as we spend most of our time indoors. The quality of the indoor environment mostly depends on proper design and precise control of ventilation systems, which require knowledge of indoor air dynamics. Building air dynamics is usually a nonlinear, transient process, which depends on many factors such as weather, building geometry, occupants' behavior, etc. Thus, indoor air distribution usually involves various complex physical phenomena, such as natural and forced convection. As a result, the task of precise prediction of indoor air parameters is not straightforward.

Computational fluid dynamics (CFD) is a very powerful tool for evaluating indoor air distribution. In CFD, the physical domain is divided into a finite number of control volumes to solve the Navier–Stokes equations numerically. The solution provides a complete set of air parameters for each control volume. However,

despite the sustained growth in computational power and advances in numerical algorithms, accurate CFD simulations are still prohibitively expensive [1] for most of the practical building applications and are mainly used for research and particular high-end buildings. The usual alternatives to CFD are the multi-zone [2] and zonal models [3], which can provide very rapid predictions but offer limited information due to the assumptions required. Moreover, the growth of computational resources in the foreseeable future would not be enough to make CFD available for routine use in building applications [1]. As a result, new numerical models capable of providing accuracy comparable with high-fidelity CFD but at considerably lower computational cost are needed.

Over the last years, several attempts to develop an alternative to zonal and multizone reduced-order models for building simulations have been made. For example, Li et al. [4] investigated a multiple model approach for predictive control of indoor thermal environment using proper orthogonal decomposition (POD). Phan et al. [5] developed a reduced-order model of a data center with multi-parameters using the POD method. Wang et al. [6] used CFD-based POD model to optimize air supply schemes in indoor environments. These models successfully reduce the simulation time while maintaining an acceptable level of accuracy. However,

Abbreviations: CFD, Computational fluid dynamics; ML, Machine learning.

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these works focus on temperature and thermal load predictions and do not consider motion-related flow parameters, which are usually more complex as they are described by nonlinear processes. Zuo & Chen [7] applied fast fluid dynamics (FFD) [8] method for indoor environmental simulations. They have successfully reduced the computing cost compared to CFD but at the expense of a considerably lower accuracy.

Data-driven models (DDMs) are steadily gaining popularity in building modeling applications due to their accurate approximations of nonlinear processes. They are used as surrogate models for building design, sensitivity and uncertainty analysis, thermal load predictions, and model predictive control (MPC) [9]. For instance, DDM-MPC for heating, ventilation, and air conditioning (HVAC) systems were developed for a university building [10], an airport [11], and a residential building [12,13]. DDMs for thermal load prediction were developed for a single building [14] and for a non-residential district [15]. On the other hand, Park & Park [16] performed a comparative analysis on the predictability of natural ventilation rates. However, these models are not based on CFD simulations and rely on reduced-order models, experimental results, or historical data to generate the database for their DDMs.

Using CFD simulations for surrogate modeling usually results in the high computational cost of dataset generation, yet a considerable number of researchers use this approach [17]. For example, Athavale et al. [18], and Fang et al. [19] compared different CFD-based DDSM approaches for temperature prediction in data centers and concluded that results produced by these models are in good agreement with the reference CFD data. Waley et al. [20] created an accurate model of vehicle cabin thermal comfort prediction using machine learning (ML) algorithms and high-fidelity CFD simulation results. The models mentioned above are of practical significance. Nevertheless, they are developed for specific applications and cannot be directly used to predict flow parameters in a more generic indoor environment. Zhou & Ooka [21] studied the influence of data preprocessing on DDM performance for reproducing CFD simulations of indoor airflow distribution and concluded that without data preprocessing, DDMs are likely to result in poor predictions. However, the objective of their work was not to reduce the simulation cost by using surrogate modeling. On the other hand, Lamberti & Gorlé [22] tackled the problem of the high computational cost of CFD database generation by combining low and high fidelity CFD simulations in an ML model to predict wind loads on buildings. Tian et al. [23] used ML to model indoor environment indicators in the stratified environments. All the researchers mentioned above note that DDMs produce accurate predictions at a low computational cost and could be an alternative to classical modeling techniques.

DDMs are based on using data analysis to find relations between system state variables without explicit knowledge of the physical behavior of the system. They can be developed relatively easily since they do not require an understanding of system physics. A comprehensive set of the high-quality input–output dataset is needed to train these models for all possible working conditions. The accuracy of DDMs decreases when training data deviates from testing data. Therefore, it is critical to use training data covering all the operating conditions, which could be challenging. However, difficulties in obtaining high-fidelity training data are compensated by the high accuracy and the low computational cost of the resulting model [12]. Therefore, DDMs could be used for complex indoor environments with stratification, natural, and forced convection, where multizone and zonal reduced-order models cannot be relied on, and computationally expensive CFD simulations are required.

In this work, we develop a data-driven surrogate model (DDSM), which is based on the data from CFD simulations. The model predicts comfort-related airflow parameters in a ventilated

room with a heated floor, which is a classical benchmark test case used by many researchers in the field of indoor environmental simulations [24–27] for testing of novel simulation methods. We have chosen this test configuration because of the challenges it presents (the mixture of natural and forced convection) and because of the availability of both experimental [28] and direct numerical simulation (DNS) CFD [1] data for validation. Our work aims to develop a model which acts as a proof of concept. The main focus of our research is on investigating the capabilities and limitations of this model as a fast and robust alternative to CFD, taking into account specific requirements for indoor environmental applications. We begin with testing different ML methods to choose the approach that best fits our model specifics. After selecting the appropriate ML method, we optimize our DDSM to accommodate the requirements of indoor environment control applications. The optimization consists of two steps. First, we change the structure of the input data so that it takes the values of temperature and velocity in the locations, which in a practical case could be replaced by sensor readings. Then we test different combinations of input probe positions in order to find the optimal sensor placement.

The content of this paper is organized as follows: Section 2 describes the governing equations, details of the test case, and the database generation; Section 3 describes the numerical details of the CFD simulations and ML methods used in this study; Section 4 shows the results of the comparison of different DDSMs and the analysis of the optimal thermostat placement; Section 5 provides a discussion on model advantages and disadvantages; and Section 6 contains concluding remarks.

2. Physical problem and dataset generation

This section describes the physical details of the simulated test case, the governing equations of the flow, and the details of the generated dataset.

2.1. Governing equations

We use the incompressible Navier–Stokes equations for a Newtonian fluid with constant physical properties. We adopt the Boussinesq approximation to account for the density variations due to temperature differences. Thermal radiation is neglected. Under these assumptions, the governing equations are

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \nu \nabla^2 \mathbf{u} - \nabla p + \beta g \Delta T, \quad (2)$$

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \alpha \nabla^2 T, \quad (3)$$

where $\mathbf{u} = (u, v, w)$ is the velocity vector in Cartesian coordinates $\mathbf{x} = (x, y, z)$, p the kinematic pressure, T the temperature, ν the kinematic viscosity, $\mathbf{g}$ the gravitational acceleration, β the thermal expansion coefficient and α the thermal diffusivity.

Hereafter, all the results are presented in dimensionless form. The reference values of time, velocity, temperature, and length are $t_{ref} = H/U_{ref}$, $U_{ref} = U_{in}$, ΔT , and H , respectively, where H is the cavity height, U_{in} is the inlet bulk velocity, $\Delta T = T_h - T_c$ is the temperature difference, T_h is the temperature of the hot wall, and T_c is the temperature of the cold wall.

2.2. Physical problem

The physical setup used in this work is a classical benchmark case of a three-dimensional ventilated cavity with a heated floor. This configuration was first studied experimentally by Blay et al. [28]. In our previous work [1] we conducted high-fidelity DNS

simulations of this benchmark case in order to validate our CFD results. This flow configuration presents a mixed convection phenomenon, challenging due to the interaction of both natural and forced convection. It resembles an airflow in a middle section of a room with mixing ventilation and thermal exhausts. The geometry of the studied cavity is shown in Fig. 1 (left). Cold air at $T_c = -0.5$ enters the cavity through the long thin inlet at the top of the left wall. The inlet velocity profile in the vertical (y) direction corresponds to a parabolic Poiseuille flow. The inlet slot has an aspect ratio $A_{in} = h_{in}/H = 0.018/1.04$. The air is discharged through the outlet slot with the aspect ratio $A_{out} = h_{out}/H = 0.024/1.04$ at the bottom of the right wall of the cavity. The bottom wall is maintained at a hot temperature of T_h , while the three other sidewalls are kept at the cold temperature of T_c .

The cavity is filled with air ($Pr = \nu/\alpha = 0.71$). The depth aspect ratio of the cavity is $A_d = D/H = 0.3/1.04$. At the outlet, convective boundary conditions ($\partial\phi/\partial t + U_{in}\partial\phi/\partial x = 0$) are imposed for the velocity and temperature. No-slip boundary conditions are applied on the walls. The initial velocity field is set to zero and the initial temperature is set equal to the temperature at the cold wall. Periodic boundary conditions are used in the spanwise (z) direction.

2.3. Dataset generation

We build the input–output dataset by changing the width aspect ratio of the cavity ($A_w = W/H$), the Rayleigh number based on the cavity height $Ra = g\beta\Delta TH^3/(\nu\alpha)$, and the Froude number based on the ratio between the bulk inlet and buoyant velocity ($Fr = U_{in}/U_{buo} = Re_H/\sqrt{Ra}$, $Re_H = U_{in}H/\nu$ is the Reynolds number based on the cavity height). Test case configurations used in the generation of the dataset are shown in Table 1.

Chosen combinations of $A_w - Ra - Fr$ are realistic and relevant for indoor environmental applications. For example, assuming that the cavity height is 2.5 meters, the highest Rayleigh number (9.6×10^9) corresponds to a temperature difference of approximately 6 °C. On the other hand, the maximum Reynolds number based on the cavity height used in this work is 9.79×10^5 . Considering the same height of 2.5 meters, it corresponds to an inlet velocity of $\approx 1\text{ m/s}$.

2.4. Input and output parameters

As input parameters of our DDSM, we initially consider Froude number (Fr), Rayleigh number (Ra), cavity width aspect ratio (A_w), temperature (T) and velocity magnitude (V) probes at seven differ-

ent locations on the mid-depth cavity plane ($z = D/2$). In total, we use 17 ($Fr + Ra + A_w + 7T + 7V$) input parameters. The positions of the probes are shown in the Fig. 1 (right). The positions of the probes are chosen according to the results of our previous work [29]. Most probes are located near the walls of the cavity in order to mimic the positions of real temperature and velocity sensors. The input dataset with all of the seven probes is used to compare the performance of different ML methods. For the problem of finding the optimal sensor location, the number of input features is reduced, which is explained in detail in Section 4.

As the outputs of the model, we choose five global flow parameters: average Nusselt number on the hot wall – $\langle Nu \rangle$, average kinetic energy – $\langle E \rangle$, average enstrophy – $\langle \Omega \rangle$, average temperature of the cavity – $\langle T_V \rangle$, and the jet separation point – x_{sep} . They represent basic airflow properties and are relevant for thermal comfort [30]. The average Nusselt number is a measure of heat transfer. It is computed using the temperature gradient at the bottom wall surface:

$$\langle Nu \rangle = -\frac{1}{WD} \int_{x=0}^W \int_{z=0}^D \frac{\partial \langle T \rangle}{\partial y} dx dz \text{ at } y = 0, \quad (4)$$

where, the standard bracket “ $\langle \rangle$ ” notation is used for time-averaged values. The average temperature is the operative room temperature. It represents the thermal properties of the flow and is averaged over time and cavity volume:

$$\langle T_V \rangle = \frac{1}{WHD} \int_{x=0}^W \int_{y=0}^H \int_{z=0}^D \langle T \rangle dx dy dz. \quad (5)$$

Kinetic energy is a measure of the level of motion, whereas enstrophy is a measure of turbulence intensity. Both kinetic energy and enstrophy are directly related to draught and local discomfort. They are averaged over time and cavity volume and calculated as follows:

$$\langle E \rangle = \frac{1}{WHD} \int_{x=0}^W \int_{y=0}^H \int_{z=0}^D \frac{\langle \mathbf{u}^2 \rangle}{2} dx dy dz \quad (6)$$

$$\langle \Omega \rangle = \frac{1}{WHD} \int_{x=0}^W \int_{y=0}^H \int_{z=0}^D \langle \omega^2 \rangle dx dy dz, \quad (7)$$

where $\omega = \nabla \times \mathbf{u}$ is the vorticity. Jet separation point is an important parameter to measure velocity comfort in a room. It is determined as a point at the top cavity wall, where the wall-shear stress $\langle \tau_w \rangle$ is equal to zero:

$$x_{sep} = x, \text{ at } \langle \tau_w \rangle = \int \frac{\partial \langle u \rangle}{\partial y} dz = 0, y = H. \quad (8)$$

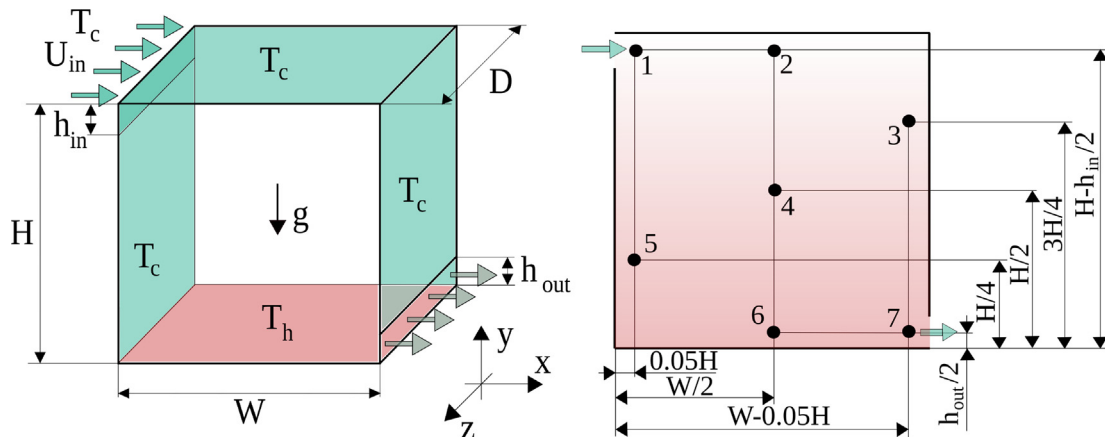


Fig. 1. Left: geometry of the studied test case. Right: locations of the input data probes at the mid-depth cavity plane ($z = D/2$).

Table 1

Combinations of the test case configurations for generating the CFD dataset. “CG” stands for coarse-grid “FG” for fine-grid CFD simulations. Each combination for A_w and Ra is tested for all of the 20 Fr numbers listed in the table.

A_w	Ra			
	1.5×10^8	6.0×10^8	2.4×10^9	9.6×10^9
0.25	FG	FG	FG	CG
0.50	FG	FG	FG	CG
1.00	FG	FG	FG	CG
2.00	FG	FG	FG	–
4.00	CG	CG	CG	–
$Fr = 0.15, 0.20, 0.25, \dots, 0.55, 0.60, 0.70, \dots, 1.50, 1.60$				
Total number of coarse-grid (CG) simulations				120
Total number of fine-grid (FG) simulations				240
Total number of simulations				360

3. Numerical methods

3.1. CFD simulations

To generate input and output data for the model, we use large-eddy simulations (LES) on staggered grids with second-order symmetry-preserving spatial discretization [31] and a one-parameter fully explicit second-order temporal discretization scheme [32]. To perform the simulations, we use an in-house CFD code developed by Gorobets et al. [33] with the LES-S3PQ turbulence model [34]. The choice of the turbulence model, type of spatial and temporal discretizations, and CFD software are based on the findings of our previous work [1]. There we performed an extensive validation and mesh sensitivity analysis of the same test case ($Ra = 2.4 \times 10^9$, $Fr = 0.81$, and $A_w = 1$) comparing different LES and Reynolds average Navier–Stokes (RANS) turbulence models, spatial grid resolutions and discretization techniques. The Cartesian staggered grid with $N_x = 120$, $N_y = N_{outlet} + N_{bulk} + N_{inlet} = 20 + 120 + 20 = 160$, $N_z = 30$ or 5.76×10^5 total control volumes showed the best trade-off between computational cost and accuracy for the aforementioned test configuration. This grid resolution is used as reference for the fine grids in this study. On the other hand, the coarse grid resolution is based on the grid with $N_x = 60$, $N_y = N_{outlet} + N_{bulk} + N_{inlet} = 7 + 60 + 8 = 75$, $N_z = 16$, which is the coarsest grid that showed reasonable accuracy in our previous study [1].

The task of choosing optimal grid discretization for the CFD simulations is a complex procedure, especially if one plans to simultaneously perform many simulations. On one hand, the nature of LES turbulence modeling requires sufficient spatial discretization [35] in order to obtain accurate simulation results. On the other hand, the high computational cost of the simulations requires the optimization of the grid size by tuning it to the physics and geometry of each individual test case. Moreover, it is not feasible to conduct mesh sensitivity analysis and validation for hundreds of CFD simulations. As a solution to this problem, we have developed an algorithm for the automatic mesh generation based on the variable physical parameters of the experiment (Fr , Ra , A_w), previously conducted validations [1], and desired level of mesh refinement.

All of the computational grids are structured and Cartesian. The schematic image of a computational grid used in this study is shown in Fig. 2. The biggest cells are located in the center of the domain, their maximum size in the vertical (y) and horizontal (x) directions ($\Delta x_{max} = \Delta y_{max}$) is determined using the Grötzsch estimate [36] defined as follows:

$$\eta_{Gro} \leq \frac{\pi Pr^{1/2}}{(Nu - 1)Ra^{1/4}} \quad \text{for } Pr \leq 1 \quad (9)$$

$$\Delta y_{max} \approx C_r \eta_{Gro}, \quad (10)$$

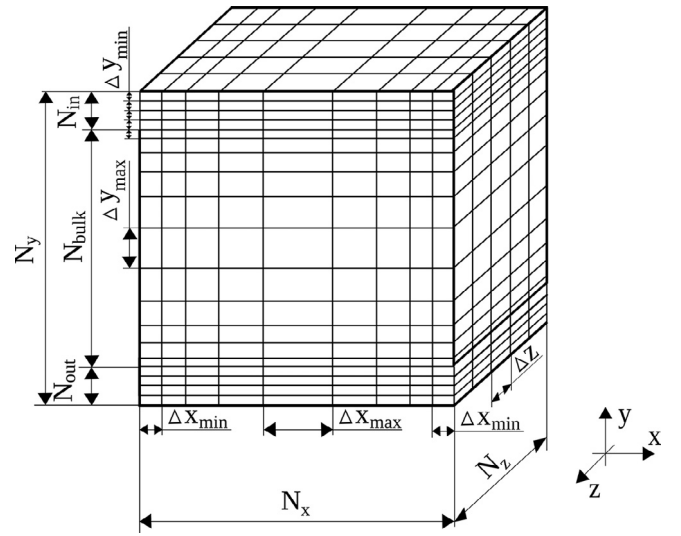


Fig. 2. Schematic image of the computational grid used in the study.

where the average Nusselt number Nu is approximated using the expression $Nu \approx 0.373(FrRa)^{0.2}$. The expression is derived using results of our previous works [37,29].

The smallest cells are located at the walls, their minimum cell size in the vertical direction – Δy_{min} is determined by the flow in the inlet area and calculated as:

$$\Delta y_{min} \approx C_r \frac{h_{in}}{2} Re_{\tau}^{-1}, \quad (11)$$

where $Re_{\tau} = \sqrt{3/2} Re_{h_{in}}$ is the friction Reynolds number for the laminar flow, and $Re_{h_{in}} = U_{in} h_{in} / \nu$ is the Reynolds based on the inlet height. C_r is the coefficient of the refinement. Eqs. (9)–(11) estimate the size of the computational grid required for the DNS simulations; thus, the value of $C_r = 1$ corresponds to the DNS level of refinement. The use of LES turbulence modeling allows the use of coarser grids, thus values of $C_r > 1$ are adopted in this study.

In the horizontal (x) direction grids are refined near the walls using the hyperbolic tangent function:

$$x = \frac{W}{2} \left(1 + \frac{\tanh\{\gamma_x(2(i-1)/N_x - 1)\}}{\tanh \gamma_x} \right), \quad (12)$$

where, γ_x is the concentration factor, N_x is the number of grid points, and $i \in [1, 2, \dots, N_x]$ is the index of the grid point in the horizontal direction respectively. Their values follow by imposing $\Delta x_{min} = 4\Delta y_{min}$ and $\Delta x_{max} = \Delta y_{max}$, where Δx_{min} is the first off-wall control volume, Δx_{max} is the control volume in the center of the cavity, and $\Delta x_{min}/\Delta y_{min} = 4$ is the ratio, used to balance the trade-off between the computational cost and accuracy.

In the vertical (y) direction the grids are uniform in the zones of the inlet ($N_{in} = h_{in}/\Delta y_{min}$) and the outlet ($N_{out} = N_{in}$) and refined in the bulk region using a hyperbolic tangent function similarly to Eq. (12). Both γ_y and N_{bulk} are found iteratively using the ratio between the minimum, Δy_{min} and the maximum, Δy_{max} cell size. The grids are uniform in the spanwise (z) direction and the number of grid points is calculated as $N_z = 1.1D/\Delta y_{max}$, where 1.1 is the ratio suggested by [38].

However, the level of grid refinement should be high enough, in order to ensure grid independence. Unfortunately, for the sake of covering the maximum amount of the possible working conditions using limited computational resources, some of the CFD simulations were performed using coarse computational. Table 1 shows the level of grid refinement for each batch of the test cases. For the fine grids, we used $C_r = 3$ (27 times coarser than DNS), and for the coarse grid simulations, we used $C_r = 6$ (216 times coarser than DNS). According to our previous studies [1], where we performed and validated CFD simulations of the same test case, the computational grid with $C_r = 3$ is sufficiently fine to achieve accurate results.

All simulations run for 500 non-dimensional time units (H/U_{in}), a long enough time-integration period to record the flow statistics for further averaging. In total, we carry out 360 CFD simulations (see Table 1), 20% of which are reserved for testing, and 80% are used in the model training. Test set is chosen randomly from the whole dataset before the beginning of the model training process.

3.2. Data-driven models

This section gives a brief description of the machine learning approaches used in this study together with the description of the training and testing process. We used four different DDSMs, namely artificial neural network (ANN), support vector regression (SVR), gradient boosting regression (GBR), and Gaussian process regression (GPR).

ANN is a machine learning framework where the input is mapped to the output using a set of interconnected nodes or neurons. Neural networks can implicitly detect complex non-linear relationships between dependent and independent variables and support multiple training algorithms. However, their disadvantages include its “black box” nature and proneness to overfitting. Usually, ANN consists of one input layer, one or several hidden layers, and one output layer. The number of neurons in the input and output layer represents the number of model parameters and output values. In contrast, the number of neurons in the hidden layer can be arbitrary. Training neural networks involve determining the appropriate combination of the number of hidden layers, the number of neurons in them, and the associated weight coefficients that minimize the prediction error. The ANN that showed the minimal prediction error is composed of one hidden layer with 16 neurons and a rectified linear unit (ReLU) neuron activation function. All of the tested hyperparameter combinations are summarized in Table 2. In our work, we use the open-source ANN library Keras [39].

SVR [40] is a machine learning framework that encodes non-linear relationships between the inputs and outputs by mapping the data in a high dimensional feature space, where the number of dimensions corresponds to the number of inputs. SVR is a robust model which can effectively work with a small number of training samples. However, SVR is not suitable for large datasets and does not perform well on noisy datasets. Training SVR is essentially performing linear regression in that high dimensional feature space. The mapping function is also called the “kernel” function. During the optimization of our SVR model, we tested different kernels, regularization strength values, and the sizes of epsilon-tube. Regu-

Table 2

Summary of the tested machine learning approaches and their hyperparameters.

Model name	Tested hyperparameters	Values	Final set
ANN	Hidden layers	1, 2	1
	Hidden layer neurons	2, 4, ..., 20	16
	Activation function	Linear, ReLU	ReLU
SVR	Kernel	Linear	RB
		Polynomial	
	Regularization strength	Radial basis (RB)	100
		10, 50, ..., 500	
GBR	Epsilon-tube width	0.01, 0.02, ..., 0.1	0.05
	Learning rate	0.01, 0.02, ..., 0.1	0.02
	Estimators	10, 50, ..., 500	100
	Max tree depth	2, 3, ..., 8	3
	Minimal samples split	2, 3, ..., 8	4
	Minimal samples leaf	2, 3, ..., 8	2
GPR	Kernel	Linear	RB
		Dot product	
		Radial basis (RB)	

larization strength is a parameter which discourages the creation of complex predictive models and controls overfitting. The size of epsilon-tube determines the width of the tube around the estimated function. Points that fall inside this tube are considered as correct predictions and are not penalized by the algorithm. The optimal hyperparameter configuration for our SVR model is the radial basis kernel function with the values of regularization strength and epsilon-tube size equal to 100 and 0.05, respectively. Other tested configurations are summarized in Table 2.

GBR [41] is a technique that combines weak prediction models (usually decision trees) into a single strong learner in an iterative fashion. The algorithm adds one tree at each stage, optimizing a cost function by moving in the negative gradient direction. GBR usually shows high accuracy prediction and fast training. Nevertheless, this model, similarly to ANN, is prone to overfitting. In our GBR model, we use regularization, characterized by the learning rate. The decision trees in GBR are characterized by the maximum tree depth, a minimal amount of samples to split a brunch, and a minimal amount of samples to form a leaf. Another critical parameter of GBR is the number of estimators, which is the number of boosting stages to perform. We tested different values of the aforementioned parameters (see Table 2). We chose GBR with the learning rate of 0.02, 100 estimators, maximum tree depth of 3, minimal samples split of 4, and minimal samples leaf of 2. This combination of parameters showed the highest accuracy for our model.

GPR [42] is a kernel-based machine learning technique for non-linear regression problems, similar to SVR. A Gaussian process (GP) is a set of random variables, such that any finite subset of these variables has a joint Gaussian distribution. GPR is a flexible approach capable of adapting to a wide range of problems by customizing kernel functions. Yet, GPR usually loses efficiency in high dimensional spaces with many input parameters and cannot handle large datasets. As a distribution, a GP is characterized by its kernel function and covariance function. Due to the Bayesian context of its formulation and interpretation, GPR has a probabilistic nature and gives prediction intervals instead of specific prediction points. In our work, we use radial basis kernel function with length scale equal to 0.2 and length scale bounds of (10^{-3} , 10^3) and the Gaussian squared-exponential covariance function, since they showed the highest prediction accuracy on our data. However, we also tested dot product kernel and linear kernel functions.

For the SVR, GBR, and GPR frameworks, we use the open-source library Scikit-learn [43]. The nature of these three models (SVR, GBR, and GPR) only supports single output problems. In order to predict all five output parameters in a single model, we use a multi-output regressor, which fits one regressor per target.

3.3. Data preprocessing and metrics

One of the most critical aspects of creating DDSMs is data preprocessing. The available dataset should be adequately scaled to make sure all the features have the same order of magnitude. In our work, we normalize all the input and output data to fit the range of $[-1, 1]$. Similar input parameters are grouped in order to apply the same normalization scale; for example, all the values of the velocity magnitudes from all the sensors and all the scenarios are normalized simultaneously. The same procedure is done for all of the temperature sensor readings. However, each of the output parameters is scaled separately, as well as the Fr , Ra and A_w from the input dataset.

We use 80% of the available data for model training and 20% for testing. Moreover, to improve the prediction results and avoid model overfitting (lack of generalization) we use cross-validation. In this study, we adopt a leave-one-out (LOO) cross-validation method. LOO is a specific case of the standard k-fold cross-validation [44]. For the LOO cross-validation at each training run, the learning set is created by taking all the samples except one, the validation set being the sample left out. Thus, for N samples, we have N different training sets and N different validation sets. This cross-validation procedure maximises the amount of data used for training since only one sample is removed from the training set. Moreover, LOO cross-validation is less biased because of the small difference in size between the training set used in each fold and the entire dataset. It is a suitable technique for small datasets.

In order to quantify the accuracy of the model, we use mean relative prediction error (MRE), which is designed as follows:

$$MRE(\phi) = \frac{1}{N} \sum_{i=1}^N \frac{|\phi_{CFD} - \phi_{DDSM}|}{|\phi_{CFD}|}, \quad (13)$$

where N is the number of samples in the test dataset, ϕ_{CFD} represents any one of the 5 comfort-related parameters calculated from the CFD simulations, and ϕ_{DDSM} represents a prediction from one of the DDSMs of any one of the 5 comfort-related parameters. We assume that less than 5% MRE is acceptable for this model.

4. Results

4.1. Comparison of different DDSM approaches

Table 3 shows a comparative analysis of results for all the models developed in this study. The table presents MREs (Eq. (13)) for each of the model output parameters (Eqs. (4)–(8)). The result are obtained using 17 ($Fr + Ra + A_w + 7T + 7V$) input parameters from 360 CFD simulations, among which 288 were used for training and 72 – for testing. Nusselt number on the hot wall $\langle Nu \rangle$ and average kinetic energy $\langle E \rangle$ are the flow parameters, which are accurately predicted by all of the tested models. The MRE for these two flow parameters does not exceed 5% (0.05) and is minimal for the SVR approach. However, average enstrophy $\langle \Omega \rangle$ is more difficult to be captured correctly, and ANN shows MRE as high as 18%. A similar picture could be observed for the average cavity tempera-

ture $\langle T_V \rangle$ and flow separation point x_{sep} , which is accurately predicted by all modeling frameworks except ANN. The smallest MRE for average enstrophy, average cavity temperature, and flow separation point is given by the GBR, which also has the smallest mean MRE. SVR and GPR are performing similarly to the GBR. However, ANN is not capable of producing accurate predictions for some of the output parameters.

The amount of data required for the DDSMs development is always a compromise between the computational cost and accuracy. Moreover, the creation of the dataset is usually the most time-consuming part of the model generation. CFD simulations have high computational cost, therefore it is useful to determine the optimum size of the training dataset to minimize the time and computational resources required to develop the model without compromising the model's accuracy.

Fig. 3 shows the relationships between the number of samples in the dataset and the accuracy of the prediction. The models are constructed using between 10 and 360 samples. There is a smooth decrease in the MRE with the increase of the number of samples in the dataset. Most of the models require not more than 120 samples to converge to a steady prediction error. However, ANN needs at least 200 samples to get the accurate predictions of kinetic energy and enstrophy. Further increase in the dataset size provides a negligibly small decrease of the prediction error. Nonetheless, SVR requires at least 200 samples to accurately predict the Nusselt number. Moreover, unlike other models, ANN does not always provide a smooth error decrease with the increase of the dataset size.

SVR, GBR, and GPR are performing similarly and show similar levels of prediction errors. Nonetheless, ANN performs differently from three other DDSMs; it generally exhibits lower accuracy, which is not improved by increasing the dataset size. The nature of ANNs could explain this discrepancy. Neural networks have probabilistic nature, as they fit parameters to transform the input and indirectly direct the activations of the following neurons. On the other hand, SVR and GBR are deterministic and explicitly fit parameters to direct the information flow [45]. Deterministic models usually handle small-sized datasets more accurately, while neural networks are more oriented towards complex problems involving a significant amount of data. It, therefore, appears that the GBR framework is the most suitable data-driven model for this kind of problem, as it shows an overall lowest level of errors. GBR framework is adopted for further analysis.

4.2. Optimal sensor position for the prediction of flow parameters

In this section, we adapt the previously developed model for the requirements of MPC. Predictive building models usually use sensor readings, weather data, heating and cooling loads, and geometrical parameters of the building as input data [46]. Therefore, in order to make our predictive model suitable for MPC applications, the input data should be structured in a way that once the model is trained, it would not require any additional simulations for the model usage. With this idea in mind, we decided to reduce the number of input parameters of the model and also relocate the positions of the temperature and velocity probes, so they could

Table 3
Mean relative prediction error (MRE) of the studied flow parameters for different models on the test dataset.

Model	MRE					
	$\langle Nu \rangle$	$\langle E \rangle$	$\langle \Omega \rangle$	$\langle T_V \rangle$	x_{sep}	Mean
ANN	0.012	0.022	0.184	0.164	0.219	0.120
SVR	0.010	0.004	0.032	0.034	0.079	0.032
GBR	0.018	0.025	0.028	0.017	0.064	0.030
GPR	0.034	0.032	0.095	0.016	0.065	0.048

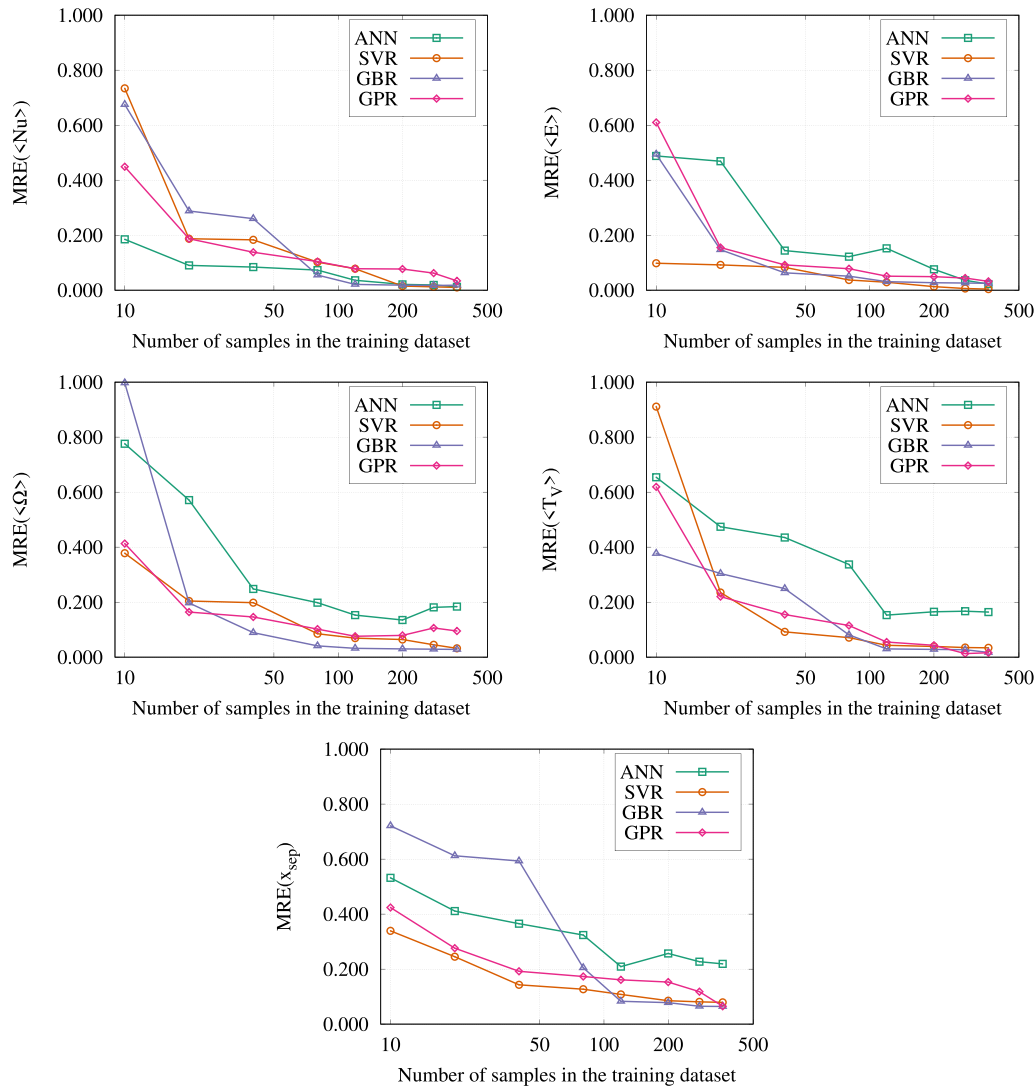


Fig. 3. Mean relative error (MRE) of the studied flow parameters for different number of samples in the training dataset and different ML models. Top left: Nusselt number on the hot wall, $\langle Nu \rangle$. Top right: average kinetic energy, $\langle E \rangle$. Middle left: average enstrophy, $\langle \Omega \rangle$. Middle right: average cavity temperature, $\langle T_V \rangle$. Bottom: flow separation point, x_{sep} .

imitate the actual thermal sensor locations. The model is then trained and tested using different combinations of input parameters in order to determine the optimal sensor position for the prediction of flow parameters.

Temperature and velocity sensors are usually located alongside the walls and ceiling of the rooms. In the previous subsection, we used the probes located as shown in the Fig. 1 (right). Not all of these probes could be replaced by the sensor readings. For example, probe 4 is located in the center of the cavity, where it is impractical to install a sensor. Likewise, probe 6 is located on the floor. In a practical case, indoor spaces are hardly ever equipped with more than two sensors per room. Thus, to optimize the prediction model's performance, the number of probes is reduced to two. Their locations are limited by the zones near the ceiling and the sidewalls (Fig. 4), which are the zones where temperature and velocity sensors could be installed. We did not consider placing more than two sensors since the model could reach sufficient accuracy with the data available. Perhaps, the increased amount of sensors could help reducing the number of samples in the training dataset; however, it is out of the scope of the present study. Moreover, in a realistic situation, apart from the cavity width aspect ratio, the values of the Froude and Rayleigh numbers are

not available; hence they are discarded from the current study. As a result, the total number of the input parameters is equal to 5 ($A_w + 2T + 2V$).

Therefore, in order to find the optimal positions of the probes, we try 361 different combinations of positions (19 per wall) for each scenario, detailed in the Fig. 4. In configuration 1, one sensor is located at the ceiling and another at the left wall; in configuration 2, one sensor is again located at the ceiling and another at the right wall. In configuration 3, there is one sensor on each lateral wall. Thick black lines in the Fig. 4 show all the possible combinations of sensors for each studied configuration, while only two positions at the time are used in the model. The position of the sensor location is considered optimal when the MRE of the predicted flow parameters is minimal. Figs. 5-9 show mean relative prediction errors of each output parameter ($\langle Nu \rangle$, $\langle E \rangle$, $\langle \Omega \rangle$, $\langle T_V \rangle$, x_{sep}) for different sensor positions for each of the studied sensor combinations from Fig. 4. The MRE for each flow parameter is averaged over the whole test set. Positions of the ceiling sensors are relative to the cavity width aspect ratio.

Fig. 5 shows the mean relative prediction error of the Nusselt number for different combinations of sensor positions. The

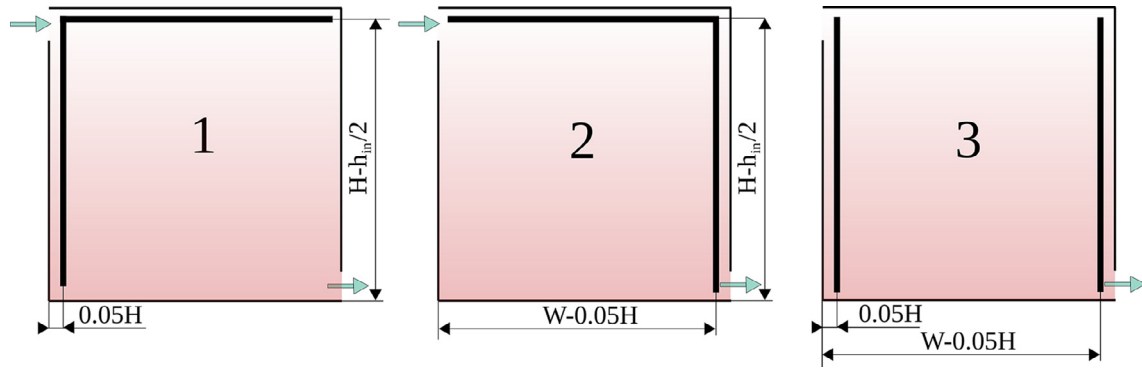


Fig. 4. Studied combinations of temperature and velocity sensor locations. Thick black lines represent the possible positions of the probes.

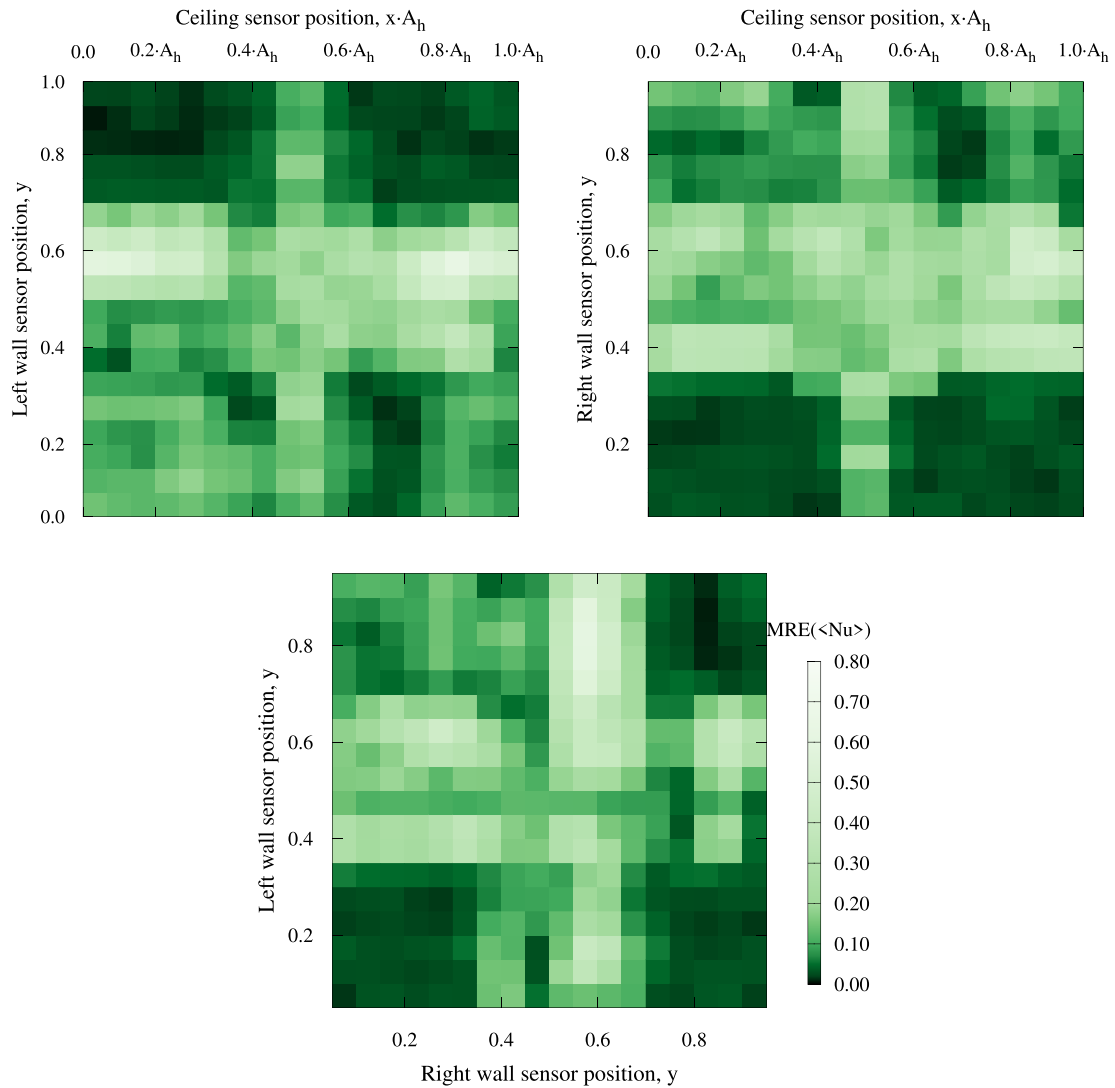


Fig. 5. Mean relative error of the averaged Nusselt number $MRE(\langle Nu \rangle)$ for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

shadows of the green show the values from the lowest MRE (dark green) to the highest (light yellow), respectively. For the configuration 1, the most accurate Nusselt number prediction is achieved when the sensors are located at the top of the left wall and near the corners of the ceiling. These are the zones where most of the

changes occur, located near the inlet and the jet separation area. For sensor configuration 2 (ceiling and the right wall), the smallest relative errors occur when the ceiling sensor is near one of the sidewalls and at the bottom of the right wall, which is close to the flow outlet. A similar tendency is observed for the third

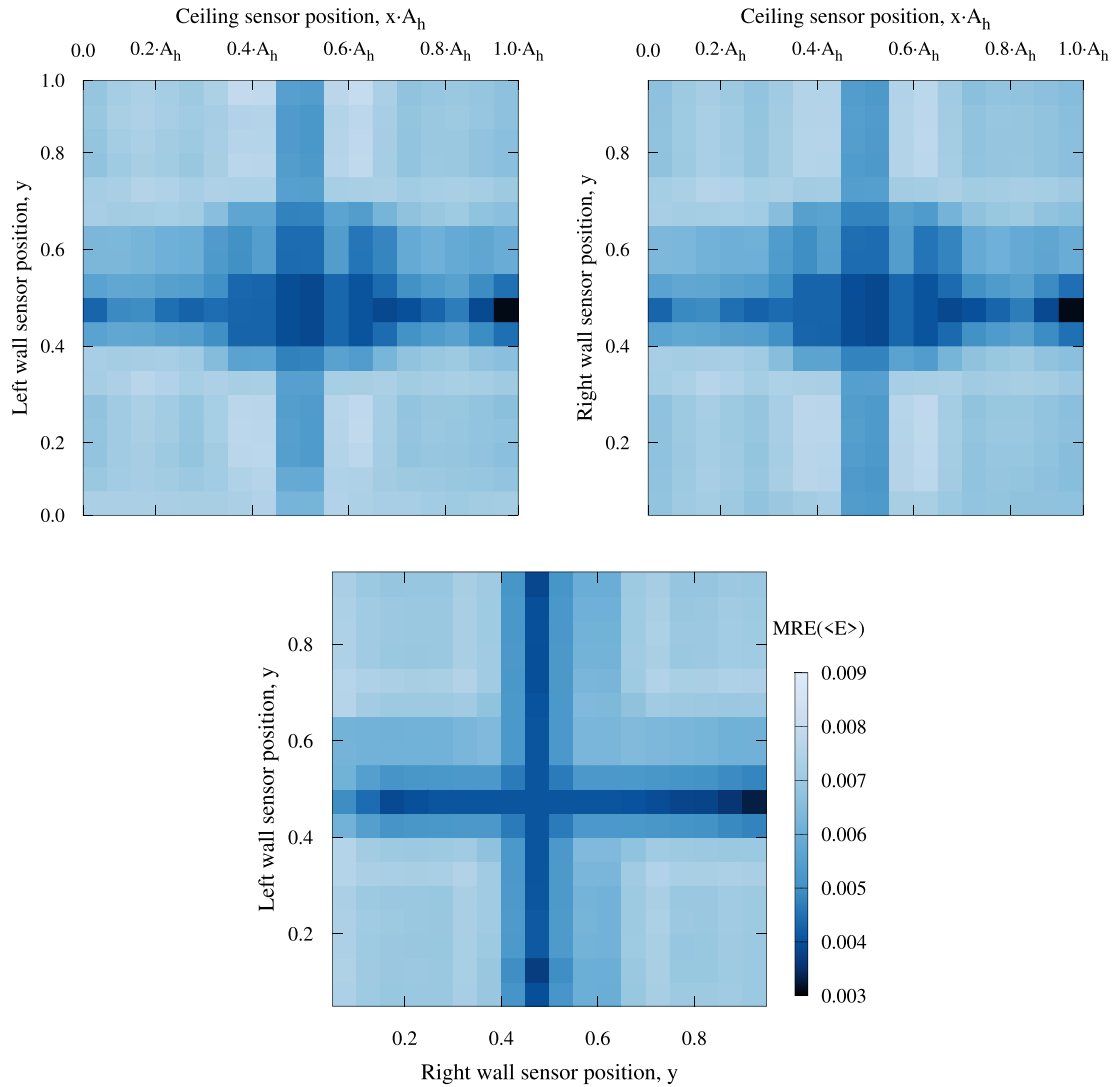


Fig. 6. Mean relative error of the averaged kinetic energy $MRE(< E >)$ for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

configuration, where sensors near the top of the left and the right wall give the most accurate predictions. Even though, the maximum MRE for the $< Nu >$ is as high as 80%, the minimum errors for all of the sensor configurations are around 1%.

As could be seen in Fig. 6, the error distribution for kinetic energy is drastically different from the Nusselt number. The minimal relative errors are located at the central areas (dark blue zones) for all sensor configurations. Moreover, the results for configurations 1 and 2 are almost identical. The best result is achieved when a ceiling sensor is placed at the right corner and the wall sensor is placed at the mid-height. For configuration 3, the error is minimal when the left wall sensor is located at the mid-height and the right wall – at the top of the wall. However, the maximum MRE does not exceed 9% for any of the studied arrangements.

Fig. 7 shows the MRE for average enstrophy. For configurations 1 and 2 (see Fig. 4), the error distribution is almost identical. The lowest errors are achieved when the ceiling sensor is placed at the corners or the center of the cavity. The wall sensor for these configurations should also be located at the top, bottom, or the wall center. On the other hand, configuration 3 shows the best results when at least one of the sensors is located in the center of the wall. However, there is a minimal zone of sensor positions,

where the $MRE(< \Omega >)$ is very high, and for most of the domain, the error does not exceed 10%.

The mean relative error for the averaged cavity temperature is shown in Fig. 8. The maximum $MRE(< T_v >)$ is reached when one sensor is placed at the central part of a ceiling and the other – at the central part of a wall (configurations 1 and 2). In this area, the error level is reaching 70%; nevertheless, the remaining sensor positions have the MRE below 5%. Configuration 3, where both sensors are placed at the sidewalls, is the most favorable for the accurate temperature prediction. Almost any combination of the sensor positions is showing good accuracy.

Flow separation point, x_{sep} , shown in Fig. 9 is easily predicted by all of the approaches, the maximum MRE does not exceed 6%. However, the error is minimal when sensors are located near the corners of the walls or the ceiling. As can be seen from the figure, it is vital to place the sensors in the area where the velocity drop occurs in order to ensure accurate prediction of the flow separation point.

The obtained results from Figs. 5–9 show that the optimal position of the sensor varies from one flow parameter to the other; thus, it is interesting to determine the points where the model gives the accurate value of all of the parameters. In order to

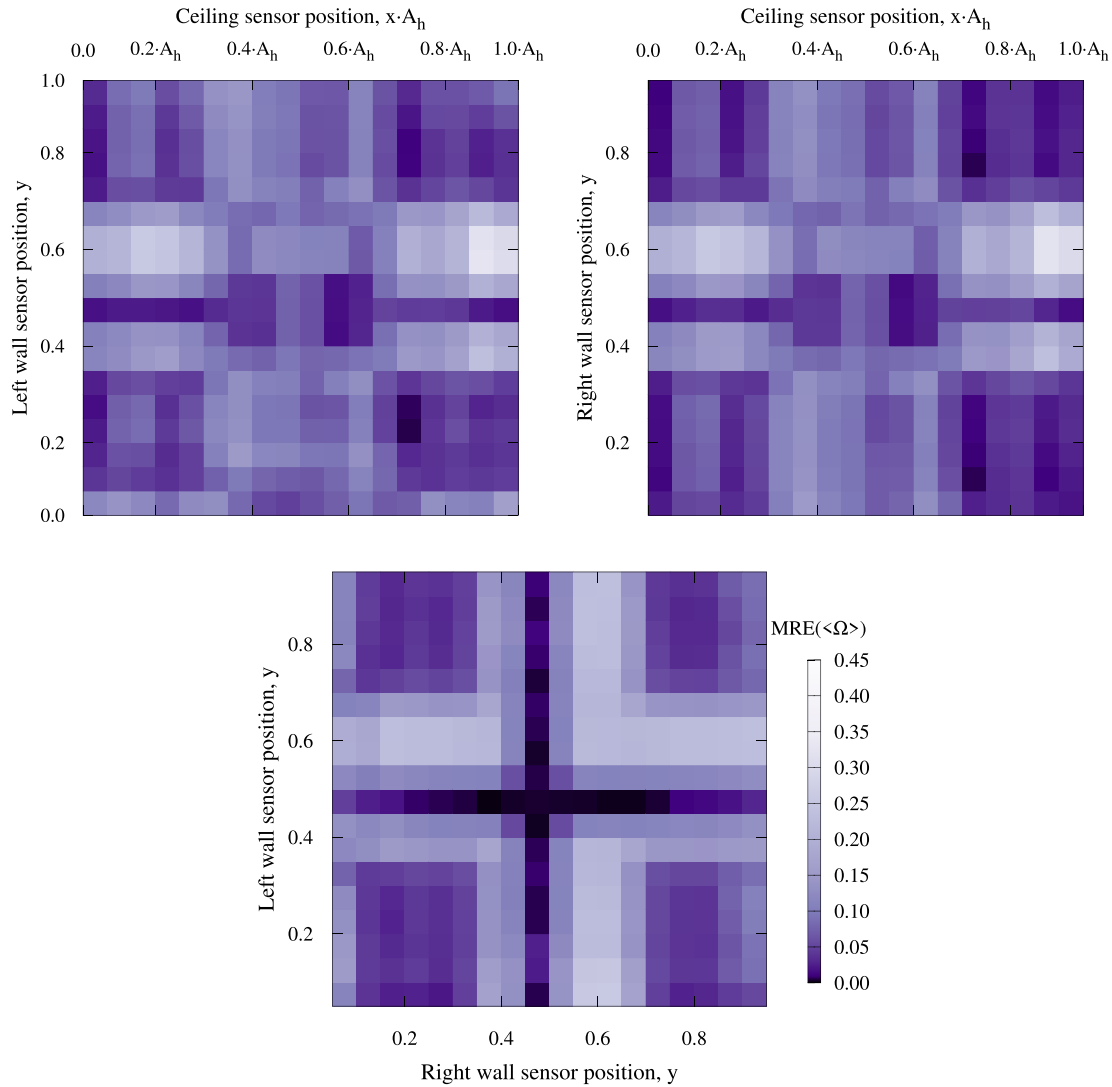


Fig. 7. Mean relative error of the averaged enstrophy $MRE(<\Omega>)$ for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

determine these points, Fig. 10 shows the maximum MRE among all of the output parameters. The scale from dark to light represents the maximum MRE from low to high, respectively. For configuration 1 (top left figure), the lowest errors are observed when the ceiling sensor is located near the corners and the left wall sensor is located near the ceiling. On the other hand, configuration 2 (top right figure) shows higher accuracy when the ceiling sensor is located near the corners, and the right wall sensor is located near the floor. Configuration 3 (bottom figure) produces the lowest MRE when the right wall sensor is located near the ceiling, and the left wall sensor is located either near the ceiling or near the floor.

It is important to note that certain sensor positions have very low sensitivity when capturing some of the flow parameters. For example, if at least one of the sensors is located in the central part of the wall, the prediction error for the Nusselt numbers increases up to 70% due to the low heat transfer in these zones. However, suppose the sensors are placed in the wall corners of the ceiling. In that case, the Nusselt number accuracy increases as a result of a more significant temperature difference in the sensor readings. On the contrary, the accuracy of the average temperature prediction benefits from sensors located in the central part of the walls

because the temperature readings in these zones are closest to the mean temperature. In contrast, sensors located in the wall corners of the ceiling introduce considerable errors due to the bigger temperature difference between them. Therefore, when placing the sensors, a decision to prioritize the accuracy of the prediction of some flow parameters over the others is to be taken into account.

In this section, we showed that the developed model is capable of producing accurate results using only the data obtained from two sensor readings. Moreover, we determined which temperature and velocity sensors positions are preferable for the studied test case. The combinations of the sensors, which achieved the smallest maximum relative error, are shown in the Fig. 11. The following section is dedicated to discussing the obtained results and the possible applications of the developed model.

5. Discussion

In this section, we present the interpretation of the obtained results, discuss the advantages and disadvantages of the data-driven models for indoor environmental simulations, and consider possible applications of the developed model.

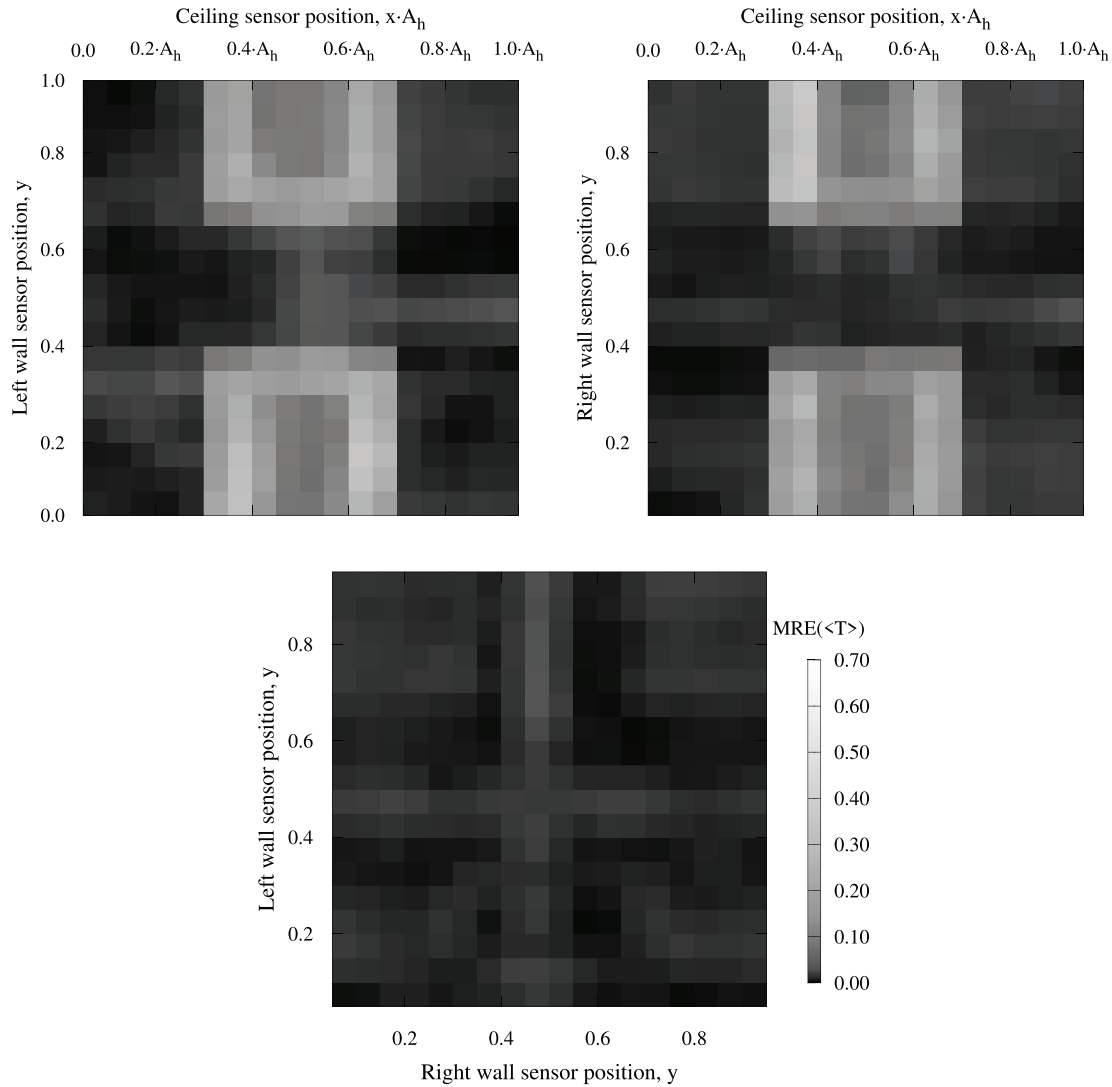


Fig. 8. Mean relative error of the averaged cavity temperature $MRE(<T_v>)$ for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

The computational cost of the model development is a combination of the computational cost of CFD simulations and the model training cost. The latter is negligible – it takes approximately 2 min to train the model on a personal computer using four CPU cores. Moreover, after the training is completed, the prediction is produced almost instantly. Virtually all the computational effort is used to produce CFD data for the model training.

The CFD database was generated using MareNostrum 4 super-computer at the Barcelona Supercomputing Center. We limited ourselves to 700Kh core-hours computational resources; hence, not all of the CFD simulations were performed with the same grid resolution (see Table 1 for details). In order to create the dataset, we were changing three physical parameters of the cases – Froude number, Rayleigh number, and the cavity width aspect ratio. The range in which these parameters were varying was chosen to represent the realistic indoor environments. However, test cases with high Rayleigh numbers and high width aspect ratios are considerably more computationally expensive than the others. We decided to compute these test cases with reduced mesh resolution in order to complete the dataset. Even though these simulations were less accurate, they still improved the accuracy of the data-driven model. The estimated cost of generating the whole dataset of 360

simulations (240 fine-grid and 120 coarse-grid) simulations is approximately 7000\$ using Amazon Web Services with the Elastic Compute Cloud Spot Instances subscription [47]. This is a justifiable computational and financial cost for the DDSM development. These CFD simulations are carried out only once to form the input dataset and will not be repeated. All of the data obtained from these simulations, including the data not used in this work, has been carefully stored for future research.

The input data of the developed model is structured so that it takes the values of temperature and velocity in the locations, which could be replaced by sensor readings. This configuration of input data makes it possible to avoid costly CFD simulations at the model execution stage. Moreover, the output parameters of the model are directly related to the parameters of the thermal comfort, such as draft rating index (DR) [48] predicted mean vote (PMV) [30], and local thermal discomfort [49]. DR can be calculated using average kinetic energy $\langle E \rangle$, average temperature $\langle T_v \rangle$, and entrophy $\langle \Omega \rangle$. The average temperature is also used in calculations of PMV, while Nusselt number $\langle Nu \rangle$ and flow separation point x_{sep} influence the local discomfort. However, the calculation of these thermal comfort coefficients is out of the scope of this work. The purpose of the developed model is a proof of

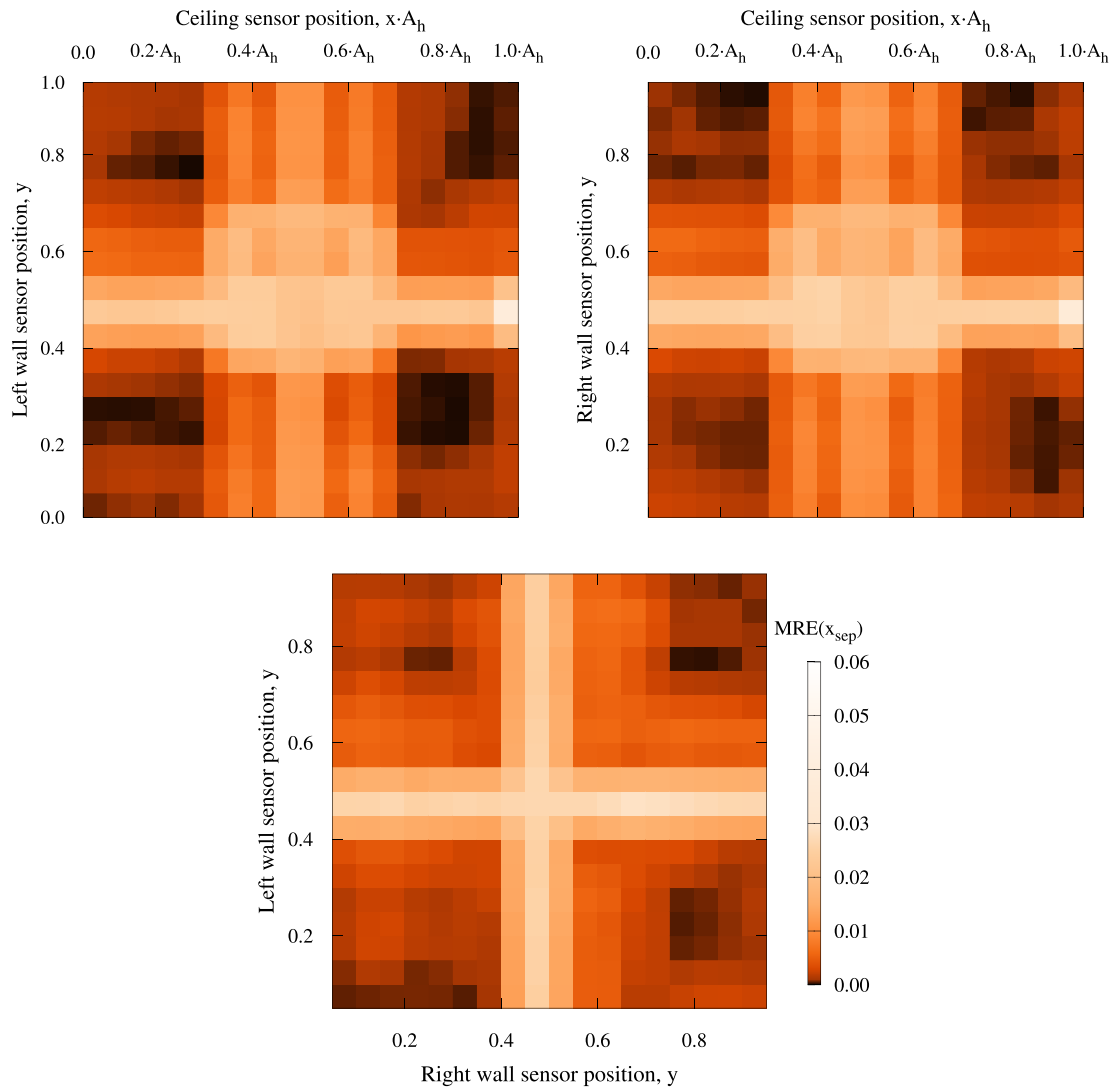


Fig. 9. Mean relative error of the flow separation point $MRE(x_{sep})$ for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

concept; thus, a classical benchmark ventilated room test case was chosen in order to increase generalization capacity and avoid problem-setup specificity. Nonetheless, once the concept is proven, the model could be extended to a direct evaluation of comfort parameters in actual rooms with specific geometries.

Despite the small number of the input parameters, the model maintains the accuracy comparable to CFD, which is a significant advantage over conventional reduced-order multizone and zonal models, which are not suitable for handling complex physical phenomena. Moreover, the small computational cost makes the model attractive for applications where fast computations are required, such as MPC or early-stage conceptual building design. However, the main bottleneck of the DDSMs is their case dependency. Even though a wide range of room geometries and physical conditions was tested, the model is not universal and does not cover a full extension of possible indoor environmental configurations. The model would not be able to produce accurate results for the case significantly different from the training dataset, for example, a room with complex geometry or completely different working conditions.

Data-driven models are capable of providing rapid predictions for common room geometries. Furthermore, as they are based on

high-fidelity data (CFD simulations), they consider temperature and velocity distributions inside individual rooms. Hence, DDSMs could be used for applications where a combination of fast and accurate predictions is required, for example, for model predictive control.

6. Conclusions and future work

In this work, we created a CFD-based DDSM for predicting the comfort-related flow parameters in a benchmark three-dimensional ventilated cavity with a heated floor. The developed DDSM provides almost instant accurate predictions using an ordinary office computer. The input parameters of the DDSM are the values of the temperature and velocity magnitude at different probe locations within the cavity domain. The output parameters are the average Nusselt number on the hot wall, the jet separation point, the average kinetic energy, the average enstrophy, and the average temperature of the cavity.

First, we created a model that uses the readings of seven probes to test different ML frameworks. The gradient boosting regression outperformed other methods, while the artificial neural network showed the least accurate results. Next, in order to imitate the

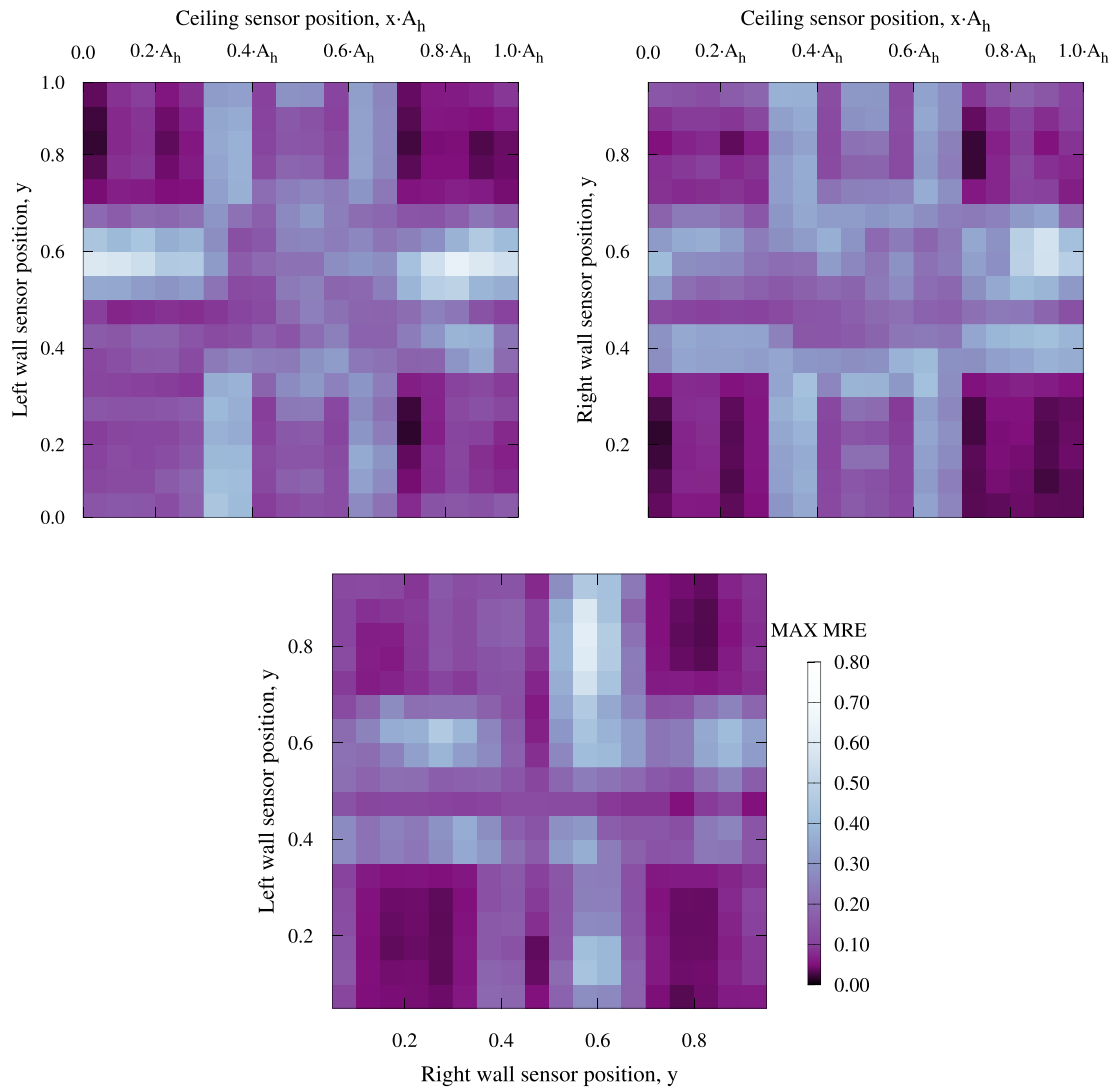


Fig. 10. Maximum (among studied flow parameters) mean relative error for different sensor positions. Top left – sensors at the ceiling and the left wall (configuration 1 in Fig. 4). Top right – sensors at the ceiling and the right wall (configuration 2 in Fig. 4). Bottom – sensors at the left and right wall (configuration 3 in Fig. 4), where the right wall is assigned to the x axis.

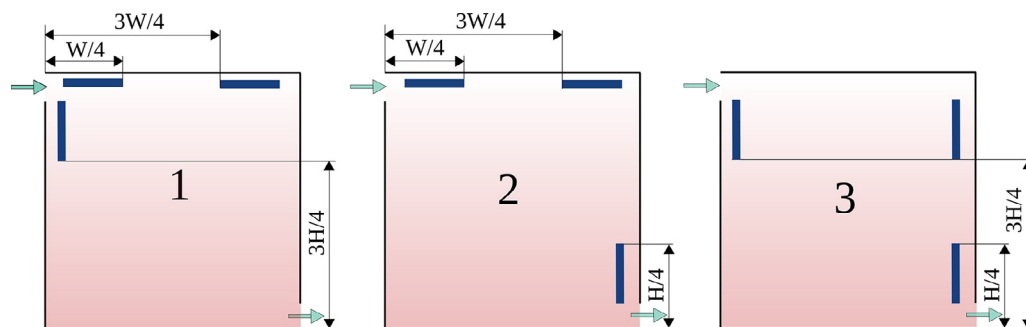


Fig. 11. Locations of temperature and velocity sensors with the highest prediction accuracy. The thick blue lines show the best sensor positions.

sensor readings, the number of probe locations was reduced to two, and the probes were placed near the ceiling and sidewalls of the cavity. Then the developed surrogate model was tested with the data from different combinations of sensor positions to determine the optimal areas of the sensor placement (Fig. 11). The model can accurately predict comfort-related flow parameters using only the information from temperature and velocity sensor

readings, which makes it possible to avoid repeated CFD simulations at the model execution stage by directly using onsite data.

The developed methodology is applicable to flow configurations with complex physical phenomena and commonly used indoor space geometrical configurations, like offices or classrooms. The methodology takes advantage of the detailed information about indoor airflow provided by accurate CFD simulations, which is

usually not accessible by multizone and zonal reduced-order models. Moreover, in the case of public buildings, for example, offices, the type of ventilation systems and office geometry does not vary significantly inside one particular building, which makes the generation of the input data easier. As a result, a comprehensive dataset could easily be created. Therefore, the developed methodology could be used for applications where fast and accurate predictions are required, for example, for ventilation operation control.

In the near future, we plan to extend the existing model for a wider range of test cases. Moreover, the information stored during the dataset generation contains time evolution statistics; thus, the model could be improved by adding the transient component.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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References

- [1] N. Morozova, F.X. Trias, R. Capdevila, C.D. Pérez-Segarra, A. Oliva, On the feasibility of affordable high-fidelity CFD simulations for indoor environment design and control, *Build. Environ.* 184 (October) (2020), <https://doi.org/10.1016/j.buildenv.2020.107144> 107144.
- [2] J. Axley, Multizone airflow modeling in buildings: History and theory, *HVAC&R Res.* 13 (6) (2007) 907–928, <https://doi.org/10.1080/10789669.2007.10391462>.
- [3] A.C. Megri, H. Fariborz, Zonal modeling for simulating indoor environment of buildings: Review, recent developments, and applications, *HVAC&R Res.* 13 (6) (2007) 887–905, <https://doi.org/10.1080/10789669.2007.10391461>.
- [4] K. Li, X. Wenping, C. Xu, H. Mao, A multiple model approach for predictive control of indoor thermal environment with high resolution, *J. Build. Performance Simul.* 11 (2) (2018) 164–178, <https://doi.org/10.1080/19401493.2017.1317289>.
- [5] L. Phan, C.-X. Lin, Reduced order modeling of a data center model with multi-Parameters, *Energy Build.* 136 (2017) 86–99, <https://doi.org/10.1016/j.enbuild.2016.11.050>.
- [6] X. Wang, J. Zhao, F. Wang, B. Song, Q. Zhang, Air supply parameter optimization of a custom nonuniform temperature field based on the POD method, *Build. Environ.* 206 (September) (2021), <https://doi.org/10.1016/j.buildenv.2021.108328> 108328.
- [7] W. Zuo, Q. Chen, Real-time or faster-than-real-time simulation of airflow in buildings, *Indoor Air* 19 (1) (2009) 33–44, <https://doi.org/10.1111/j.1600-0668.2008.00559.x>.
- [8] J. Stam, Stable fluids, in: *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques*, 1999, pp. 121–128, <https://doi.org/10.1145/311535.311548>.
- [9] P. Westermann, R. Evins, Surrogate modelling for sustainable building design – a review, *Energy Build.* 198 (2019) 170–186, <https://doi.org/10.1016/j.enbuild.2019.05.057>.
- [10] A.E. Ruano, S. Pesteh, S. Silva, H. Duarte, G. Mestre, P.M. Ferreira, H.R. Khosravani, R. Horta, The IMBPC HVAC system: a complete MBPC solution for existing HVAC systems, *Energy Build.* 120 (2016) 145–158, <https://doi.org/10.1016/j.enbuild.2016.03.043>.
- [11] H. Huang, L. Chen, E. Hu, A neural network-based multi-zone modelling approach for predictive control system design in commercial buildings, *Energy Build.* 97 (2015) 86–97, <https://doi.org/10.1016/j.enbuild.2015.03.045>.
- [12] A. Afram, F. Janabi-Sharifi, A.S. Fung, K. Raahemifar, Artificial neural network (ANN) based model predictive control (MPC) and optimization of HVAC systems: A state of the art review and case study of a residential HVAC system, *Energy Build.* 141 (2017) 96–113, <https://doi.org/10.1016/j.enbuild.2017.02.012>.
- [13] Q. Chen, N. Li, Fast simulation and high-fidelity reduced-order model of the multi-zone radiant floor system for efficient application to model predictive control, *Energy Build.* 248 (2021), <https://doi.org/10.1016/j.enbuild.2021.111210> 111210.
- [14] H. Sözer, S.S. Aldin, Predicting the indoor thermal data for heating season based on short-term measurements to calibrate the simulation set-points, *Energy Build.* 202 (2019), <https://doi.org/10.1016/j.enbuild.2019.109422>.
- [15] D. Koschwitz, E. Spinnraker, J. Frisch, C. van Treeck, Long-term urban heating load predictions based on optimized retrofit orders: A cross-scenario analysis, *Energy Build.* 208 (2020), <https://doi.org/10.1016/j.enbuild.2019.109637> 109637.
- [16] H. Park, D.Y. Park, Comparative analysis on predictability of natural ventilation rate based on machine learning algorithms, *Build. Environ.* 195 (February) (2021), <https://doi.org/10.1016/j.buildenv.2021.107744>. URL: 10.1016/j.buildenv.2021.107744 107744.
- [17] G. Calzolari, W. Liu, Deep learning to replace, improve, or aid CFD analysis in built environment applications: A review, *Build. Environ.* 206 (August) (2021), <https://doi.org/10.1016/j.buildenv.2021.108315> 108315.
- [18] J. Athavale, M. Yoda, Y. Joshi, Comparison of data driven modeling approaches for temperature prediction in data centers, *Int. J. Heat Mass Transf.* 135 (2019) 1039–1052, <https://doi.org/10.1016/j.ijheatmasstransfer.2019.02.041>.
- [19] Q. Fang, Z. Li, Y. Wang, M. Song, J. Wang, A neural-network enhanced modeling method for real-time evaluation of the temperature distribution in a data center, *Neural Comput. Appl.* 31 (12) (2019) 8379–8391, <https://doi.org/10.1007/s00521-019-04508-y>.
- [20] A. Warey, S. Kaushik, B. Khalighi, M. Cruse, G. Venkatesan, Data-driven prediction of vehicle cabin thermal comfort: using machine learning and high-fidelity simulation results, *Int. J. Heat Mass Transf.* 148 (2020), <https://doi.org/10.1016/j.ijheatmasstransfer.2019.119083> 119083.
- [21] Q. Zhou, R. Ooka, Influence of data preprocessing on neural network performance for reproducing CFD simulations of non-isothermal indoor airflow distribution, *Energy Build.* 230 (2021), <https://doi.org/10.1016/j.enbuild.2020.110525> 110525.
- [22] G. Lamberti, C. Gorié, A multi-fidelity machine learning framework to predict wind loads on buildings, *J. Wind Eng. Ind. Aerodyn.* 214 (April) (2021), <https://doi.org/10.1016/j.jweia.2021.104647> 104647.
- [23] X. Tian, Y. Cheng, Z. Lin, Modelling indoor environment indicators using artificial neural network in the stratified environments, *Build. Environ.* 208 (October 2021) (2021) 108581. doi:10.1016/j.buildenv.2021.108581. URL: 10.1016/j.buildenv.2021.108581..
- [24] Z. Zhang, W. Zhang, Z.J. Zhai, Q.Y. Chen, Evaluation of various turbulence models in predicting airflow and turbulence in enclosed environments by CFD: Part-2: Comparison with experimental data from literature, *HVAC&R Res.* 13 (6) (2007) 871–886, <https://doi.org/10.1080/10789669.2007.10391460>.
- [25] A. Limane, H. Fellouah, N. Galanis, Thermo-ventilation study by OpenFOAM of the airflow in a cavity with heated floor, *Build. Simul.* 8 (3) (2015) 271–283, <https://doi.org/10.1007/s12273-014-0205-4>.
- [26] G. Yang, Y. Huang, J. Wu, L. Zhang, G. Chen, R. Lv, A. Cai, Experimental study and numerical models assessment of turbulent mixed convection heat transfer in a vertical open cavity, *Build. Environ.* 115 (2017) 91–103, <https://doi.org/10.1016/j.buildenv.2017.01.016>.
- [27] J.Y. Wu, R.R. Lv, Y.Y. Huang, G. Yang, Flow Structure Transition and Hysteresis of Turbulent Mixed Convection Induced by a Transverse Buoyant Jet, *Int. J. Heat Mass Transf.* 177 (2021), <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121310>.
- [28] D. Blay, S. Mergui, J.L. Tuhaut, F. Penot, Experimental turbulent mixed convection created by confined buoyant wall jets, in: *Proceedings of the First European Heat Transfer Conference UK*, 1992, pp. 821–828.
- [29] N. Morozova, F.X. Trias, R. Capdevila, A. Oliva, Data-driven prediction of flow parameters in a ventilated cavity, in: *Proceedings of Building Simulation 2021: 17th Conference of IBPSA*, 2021.
- [30] Ergonomics of the thermal environment – Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria, Standard, International Organization for Standardization, Geneva, CH, 2005..
- [31] R.W.C.P. Verstappen, A.E.P. Veldman, Symmetry-preserving discretization of turbulent flow, *J. Comput. Phys.* 187 (1) (2003) 343–368, [https://doi.org/10.1016/S0021-9991\(03\)00126-8](https://doi.org/10.1016/S0021-9991(03)00126-8).
- [32] F.X. Trias, O. Lehmkuhl, A self-adaptive strategy for the time integration of Navier-Stokes equations, *Numer. Heat Transfer, Part B: Fundamentals* 60 (2) (2011) 116–134, <https://doi.org/10.1080/10407790.2011.594398>.
- [33] A. Gorobets, F.X. Trias, M. Soria, A. Oliva, A scalable parallel Poisson solver for three-dimensional problems with one periodic direction, *Comput. Fluids* 39 (3) (2010) 525–538, <https://doi.org/10.1016/j.compfluid.2009.10.005>.
- [34] F.X. Trias, D. Folch, A. Gorobets, A. Oliva, Building proper invariants for eddy-viscosity subgrid-scale models, *Phys. Fluids* 27 (6) (2015), <https://doi.org/10.1063/1.4921817>.
- [35] S.B. Pope, *Turbulent flows*, Cambridge University Press, 2000, <https://doi.org/10.1017/CBO9780511840531>.
- [36] G. Grötzbach, Spatial resolution requirements for direct numerical simulation of the Rayleigh-Bénard convection, *J. Comput. Phys.* 49 (2) (1983) 241–264, [https://doi.org/10.1016/0021-9991\(83\)90125-0](https://doi.org/10.1016/0021-9991(83)90125-0).
- [37] N. Morozova, F.X. Trias, R. Capdevila, A. Oliva, Investigating the capabilities of CFD-based data-driven models for indoor environmental design and control, in: *Proceedings of 14th WCCM & ECCOMAS Congress 2020*, 2021.

- [38] F. Dabbagh, F.X. Trias, A. Gorobets, A. Oliva, On the evolution of flow topology in turbulent Rayleigh-Bénard convection, *Phys. Fluids* 28 (11) (2016), <https://doi.org/10.1063/1.4967495>.
- [39] F. Chollet, keras, URL: <https://github.com/fchollet/keras>, 2015..
- [40] C. Cortes, V. Vapnik, Support-vector networks, *Mach. Learn.* (1995) 273–297.
- [41] T. Hastie, R. Tibshirani, J. Friedman, *The Elements of Statistical Learning*, Springer (2009), <https://doi.org/10.1007/978-0-387-84858-7>.
- [42] C.E. Rasmussen, *Gaussian Processes in Machine Learning*, Springer Berlin Heidelberg, 2004, pp. 63–71. doi:10.1007/978-3-540-28650-9_4.
- [43] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay, Scikit-learn: Machine learning in Python, *J. Mach. Learn. Res.* 12 (2011) 2825–2830, <https://doi.org/10.5555/1953048.2078195>.
- [44] M. Stone, Cross-validatory choice and assessment of statistical predictions, *J. Roy. Stat. Soc.: Ser. B (Methodol.)* 36 (2) (1974) 111–133, <https://doi.org/10.1111/j.2517-6161.1974.tb00994.x>.
- [45] S. Lundberg, G. Erion, H. Chen, A. DeGrave, J. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, S. Lee, From local explanations to global understanding with explainable AI for trees, *Nature Mach. Intell.* 2 (2020) 56–67, <https://doi.org/10.1038/s42256-019-0138-9>.
- [46] Y. Yao, D. Shekhar, State of the art review on model predictive control (MPC) in Heating Ventilation and Air-conditioning (HVAC) field, *Build. Environ.* 200 (May) (2021), <https://doi.org/10.1016/j.buildenv.2021.107952> 107952.
- [47] Amazon Web Services, Amazon EC2 On-Demand Pricing, URL: <https://aws.amazon.com/ec2/spot/pricing/> (2021)..
- [48] P.O. Fanger, A.K. Melikov, H. Hanzawa, J. Ring, Air turbulence and sensation of draught, *Energy Build.* 12 (1) (1988) 21–39, [https://doi.org/10.1016/0378-7788\(88\)90053-9](https://doi.org/10.1016/0378-7788(88)90053-9).
- [49] G.A. Ganesh, S.L. Sinha, T.N. Verma, S.K. Dewangan, Investigation of indoor environment quality and factors affecting human comfort: A critical review, *Build. Environ.* 204 (March) (2021), <https://doi.org/10.1016/j.buildenv.2021.108146> 108146.